Breeding Diverse Noise: A Population-Based Companion to "It's Never Too Late" Collapse Recovery

Anirud Aggarwal
anirud@vamana.ai research.aniaggarwal.com
Code: github.com/AniAggarwal/divgen (fork of the official implementation)

Abstract

Harrington et al.'s "It's Never Too Late" makes a lovely observation: mode collapse in trained text-to-image models is not a fact about the model so much as a fact about where we start it — and the initial noise can simply be optimized, end-to-end, into a diverse set of generations. This note is written in appreciation of that work, and asks one follow-up question of it: does the noise need a gradient to find its way? We reformulate their objective as a two-objective artificial-selection search[1] over sets of initial noises, using the multi-objective machinery from our ECAD work on caching schedules. On GenEval with SDXL-Turbo, breeding reaches approximate parity with the paper's gradient-based optimization on its own metrics — without a single backward pass through the sampler. We describe what the population view adds (memory-free scaling to large models, non-differentiable set-level objectives, an explicit quality–diversity Pareto front, self-tuned step sizes) and what it costs, and offer it as a companion, not a replacement, to the original method.

1 Appreciation, and a question

Three ideas in Harrington et al. strike us as durable. First, that inference-time diversity is recoverable at all — that a collapsed model still contains its diversity, reachable through the initial noise x_0. Second, the frequency analysis (their Fig. 9): optimization acts almost entirely on the lowest third of the noise spectrum, which both explains the method and immediately yields the pink-noise initialization that improves every method they test, including the baselines. Third, the observation that set-level objectives (DPP, Vendi) are the right currency for diversity, because they cannot be gamed by making one image strange — a point their user study makes empirically.

The method that operationalizes these ideas is gradient descent through the frozen sampler: differentiable diversity and quality scores are backpropagated to the noise batch. Our question is narrow: is the gradient essential, or is it one search procedure among several for the landscape their analysis reveals? The question matters practically: backpropagating through a 10B-parameter flow model is a memory wall on commodity hardware, and the most human-aligned objectives in their study (DPP and Vendi at the set level) are exactly the ones where differentiability is most awkward.

We answer it by transplanting the multi-objective search we built for ECAD [2] — which breeds diffusion caching schedules against a quality–speed front — onto their problem: same NSGA-II [3] engine, same two-objective structure, with the chromosome swapped from binary cache decisions to a set of noise tensors.

2 Breeding noise sets

Individuals. Diversity in the source paper is measured over a set of B=4 images per prompt, so the unit of selection is the whole set: an individual is $\{x_0^{(1)}, \dots, x_0^{(B)}\}$, plus two strategy genes (a mutation step σ and a per-element mutation rate ρ) that are inherited and perturbed alongside the noise, as in classical self-adaptive strategies [4].

Objectives. Each individual is rendered (forward passes only) and scored with the paper's own code: mean per-image CLIPScore as quality, and a set-level statistic — patchwise DINOv2 distance, DPP log-det, or Vendi — as diversity. NSGA-II maintains the Pareto front of the two, exactly as ECAD maintains quality–versus–speed. Generation 0 is drawn i.i.d. from the prior, so the paper's baseline is embedded in every run as the starting population.

[1] We deliberately avoid the customary term "evolutionary algorithm." Natural evolution optimizes no objective — it is selection without a selector, purposeless by construction — so attaching the word to an explicitly objective-driven procedure seems to us a misnomer. What this algorithm actually performs is closer to what breeders have always done: artificial selection toward a stated goal. We therefore speak of breeding noise, and of selection-based or population-based search.

Table 1: GenEval, SDXL-Turbo, white-noise initialization, set size 4. Paper rows from Harrington et al. Tab. 1 (their eval harness, 552 prompts); our rows use the official repository's own metric code over 10 prompts (preliminary; the full 552-prompt run is in progress and will replace these numbers). Diversity metrics: higher is better.

Method	DINO	DreamSim	LPIPS	CLIP
i.i.d.	0.588	0.249	0.642	0.335
Parmar et al. [5]	0.705	0.331	0.682	0.333
Harrington et al. (gradient)	0.784	0.411	0.767	0.349
Bred, gen. 0 (i.i.d.)	0.627	0.308	0.642	0.336
Bred, final	0.794	0.473	0.729	0.350

Operators, each borrowed from a finding of theirs. Mutations are spectrally colored by $1/(1+f)^{\beta}$, so proposals move low frequencies more than high ones — their Fig. 9 dynamic, made into an operator (and the same filter behind their pink-noise initialization). Their χ_d-norm regularizer $K(\epsilon)$, which keeps noises in the high-density shell of the prior, becomes a repair step: every latent is re-standardized after every operator, enforcing the constraint exactly rather than penalizing its violation. Crossover exchanges whole noise slots between parent sets — recombination in the space of sets, respecting that individual noises are the atoms of diversity.

Self-tuned exploration. In our ECAD ablations, a 1% mutation rate stalls in poor local optima and 15% converges too slowly; the sweet spot was never known. Here each individual carries its own (σ, ρ), and selection tunes them: the population raises σ early (exploration) and anneals it toward 0.05 as it converges (exploitation), with no schedule supplied by us (Fig. 1, right). A controlled comparison (Tab. 2) confirms the fixed-rate pathologies transfer to this domain and that the self-tuned variant matches the best fixed rate without knowing it in advance.

3 Results

Table 1 gives the headline. Two sanity checks first: our generation-0 population, scored with the official code, reproduces the paper's i.i.d. row almost exactly, and quality rises during search rather than being traded away. On the diversity metrics, breeding lands at approximate parity with the gradient optimizer — slightly ahead on DINO and DreamSim, slightly behind on LPIPS — despite never computing $\partial(\text{image})/\partial x_0$. We read this as evidence for a claim implicit in the original paper: the low-frequency structure of the noise landscape is benign enough that any sensible search finds it; the gradient is one excellent way, not the only one.

Table 2: Mutation-rate comparison (4 GenEval prompts, 45 generations, matched seeds), echoing our ECAD Table 9 in this new domain. The self-tuned variant needs no rate chosen in advance.

Mutation setting	DINO	DreamSim	Vendi	DPP
fixed rate 1%	0.686	0.280	1.97	0.755
fixed rate 15%	0.774	0.369	2.15	0.784
self-tuned	0.756	0.374	2.25	0.792

Qualitatively the recovery is the same phenomenon their Fig. 1 shows: four interchangeable grey park benches become a teal bench, a golden-hour park, new viewpoints — every image still plainly a bench (Fig. 2).

Two extensions the population view invites. (i) Prompt-typed islands. Sub-populations bred on disjoint prompt distributions (painted landscapes, five-word prompts, long GPT-written prompts), with the best noise sets migrating between islands every few generations. Because a noise set is prompt-agnostic, a migrant that keeps winning after crossing prompt types is one whose diversity generalizes: in our runs, migrated elites retain 0.65–0.69 DINO diversity on prompt families they never saw during selection. Per-prompt gradient descent has no natural analogue of this test. (ii) Searching program space. Instead of breeding the noise tensor, we can breed a small symbolic program (compositions of white noise, pink filtering, and low-frequency gratings) that generates it. Program search reaches the highest raw diversity we observe (DINO 0.886) at a real cost in prompt fidelity — it maps the far end of the quality–diversity front — and its early winners rediscover low-frequency structure on their own, a pleasing independent confirmation of the paper's spectral analysis.

4 What each method offers

We want to be precise about the trade, because the comparison flatters neither method alone. The gradient optimizer is far more sample-efficient where it applies: at 0.345 s per iteration on an A100 (their Sec. 4.2), it reaches its result in seconds per prompt, while breeding spends thousands of forward renders to arrive at the same neighborhood. Population search earns that cost back in three situations. Memory: it never backpropagates through the sampler, so model scale adds no optimizer memory — relevant exactly at the FLUX scale the paper targets. Objectives: non-differentiable scores — true DPP samplers, human ratings, detector-based prompt-faithfulness checks — enter the loss with no relaxation, and it is pre-

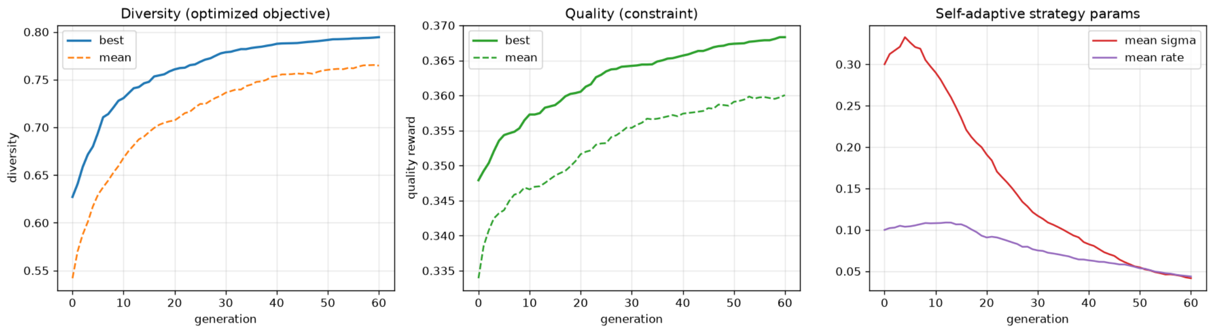

Figure 1: Convergence averaged over GenEval prompts (SDXL-Turbo, population 40). Left: the bred diversity objective. Middle: CLIP quality rises during search — the Pareto selection concedes nothing for the diversity gain. Right: the self-adaptive mutation step σ and rate ρ, raised then annealed by selection alone.

Figure 2: "A photo of a bench," SDXL-Turbo. Left: i.i.d. generation 0. Right: the bred set.

cisely such set-level scores that their user study found most human-aligned. Output: a run yields an explicit Pareto front of noise sets, so the quality–diversity trade-off becomes a menu rather than a weight to tune. The two approaches also compose naturally: bred noise is a warm start for gradient refinement, and their pink-noise initialization already improves our generation 0 the same way it improves theirs.

5 Closing

This note exists because the original paper's analysis was clear enough to build on in a week: the frequency finding told us what mutations to propose, the norm regularizer told us what constraint to repair to, and the set-level metrics told us what to select for. That is what a good paper does. We offer the population view back as a companion — same landscape, second vehicle — with code, configurations, and scaling scripts in the fork above, in the hope it is useful to the authors' future work on collapse recovery.

References

[1] A. Harrington, A. S. Koepke, S. Karthik, T. Darrell, A. A. Efros. It's Never Too Late: Noise Optimization for Collapse Recovery in Trained Diffusion Models. In CVPR, 2026.

[2] A. Aggarwal et al. ECAD: Evolutionary Caching to Accelerate Your Off-the-Shelf Diffusion Model. In ICLR, 2026.

[3] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans. Evolutionary Computation, 6(2), 2002.

[4] H.-G. Beyer, H.-P. Schwefel. Evolution strategies: a comprehensive introduction. Natural Computing, 1(1), 2002.

[5] G. Parmar et al. Scaling group inference for diverse and high-quality generation. arXiv:2508.15773, 2025.